CPE 381 Classwork 08: Continuous-Time Signals, Linearity & Superposition

Student Name (as in Canvas): _____

A Number: _____

Points: 20 Duration: 15 minutes Due: September 16, 2026 (In-Class)

1. Check if the signal $x(t) = e^{-at}u(t), \quad a > 0$ is an energy signal. Note: a signal is energy signal if its energy is finite. **(10 Points)**

Answer

2. For a system defined by $y(t) = T[x(t)]$, determine if it is additive where the system T is specified by $y(t) = x(t) + 3$. **(5 Points)**

Answer

3. Is $y(t) = x(t) + 3$ homogenous? **(5 Points)**

Answer